

DESCRIPTION

The JW[®]5712/A provides an ultra-low power solution for products powered by either one rechargeable Li-Ion batteries, Li-primary battery chemistries such as Li-MnO₂ or two to three cell alkaline batteries. The output voltage of the buck converter is set with three VSEL pins between 0.7V and 3.3V. JW5712/A features low output ripple voltage and low noise with a small output capacitor. Once the input voltage comes close to the output voltage the device enters the 100% duty cycle operation mode to prevent an increase of output ripple voltage. In this operation mode the device stops switching and turns the high-side MOSFET switch on.

JW5712/A is available in 8 ball WLCSP 1.6mmx0.9mm package, which provide a compact solution with minimal external components.

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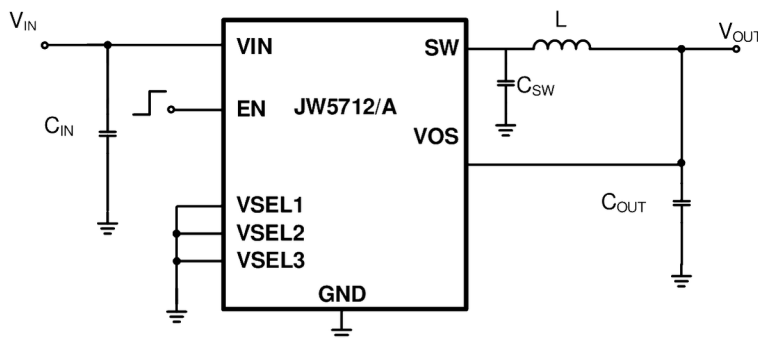
FEATURES

- 2.3V to 5.5V operating input range
- Output current
JW5712/A: 600mA, peak to 800mA
- Typical 360nA quiescent current
- Up to 90% efficiency with load current >15uA
- Low output ripple voltage
- 8 selectable output voltages
JW5712: 1.2V to 3.3V
JW5712A: 0.7V to 3.1V
- Output short protection
- Output voltage discharge
- Automatic transition to 100% duty cycle operation
- Thermal protection
- Available in WLCSP8 package

APPLICATIONS

- IOT
- Wearable and Personal Electronics
- Health Monitoring and Medical Accessories
- Industrial Metering
- Energy Harvesting

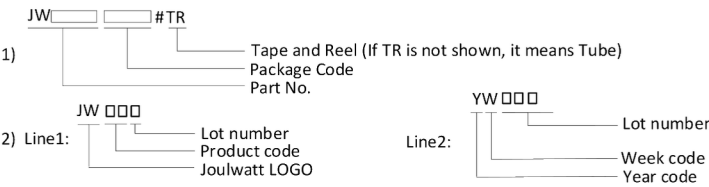
TYPICAL APPLICATION



ORDER INFORMATION

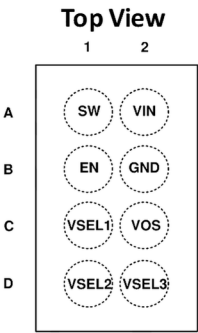
DEVICE ¹⁾	PACKAGE	TOP MARKING ²⁾	ENVIRONMENTAL ³⁾
JW5712WLCSPC#TR	WLCSP1.575x0.875-8	JWNH□ YW□□□	Green
JW5712AWLCSPC#TR	WLCSP1.575x0.875-8	JWNJ□ YW□□□	Green

Note:



3) All Joulwatt products are packaged with Pb-free and Halogen-free materials and compliant to RoHS standards.

PIN CONFIGURATION



PIN DESCRIPTION

Pin	Name	Description
A1	SW	Switch pin which is connected to the internal MOSFET switches. Connect inductor to this terminal.
A2	VIN	VIN power supply pin. Connect this pin close to the VIN terminal of the input capacitor. A minimum of 4.7 μ F ceramic capacitor from this pin to GND is required.
B1	EN	Enable pin. Drive EN pin high to turn on the regulator and low to turn off the regulator.
B2	GND	Ground pin.
C1	VSEL1	Output voltage selection pins.
D1	VSEL2	
D2	VSEL3	
C2	VOS	Feedback pin for the internal feedback divider network and regulation loop. Connect this pin directly to the output capacitor with a short trace.

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